



Professional Cloud Services

(Cloud-Modules DevOps Practitioner Challenge)

Deploy a Containerized Web App to AWS ECS

Goal

Create a complete AWS ECS deployment using Terraform and GitHub Actions.

You must deploy a simple Docker container that serves an HTML page with:

None

Hello cloud-modules

The container must listen on **port 5000**, and users must access it through an **Application Load Balancer on port 80**.

Requirements

Infrastructure

Create everything using **Terraform**:

1. VPC
 2. 2 public subnets
 3. 2 private subnets
 4. ECR repository
 5. ECS cluster
 6. Application Load Balancer
 7. Target group
 8. Security groups
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Application

Create a Docker image that contains a simple web server.

The web page must print:

None

hello cloud-modules

The container must listen on:

None
5000

GitHub Actions Pipeline

Use a **private GitHub repository**.

The pipeline must:

1. Build the Docker image
2. Authenticate to AWS securely
3. Push the image to ECR
4. Deploy the new image to ECS

Important:

Do **not** use AWS access keys or secret keys.

(Use a safer authentication method)

Acceptance Criteria

1. Terraform code for the full infrastructure
2. GitHub Actions pipeline YAML
3. Dockerfile and app files
4. Screenshot of the public route table
5. Screenshot of the private route table
6. URL of the deployed website
7. Send all results and relevant materials to: devops@cloud-modules.com

Good Luck